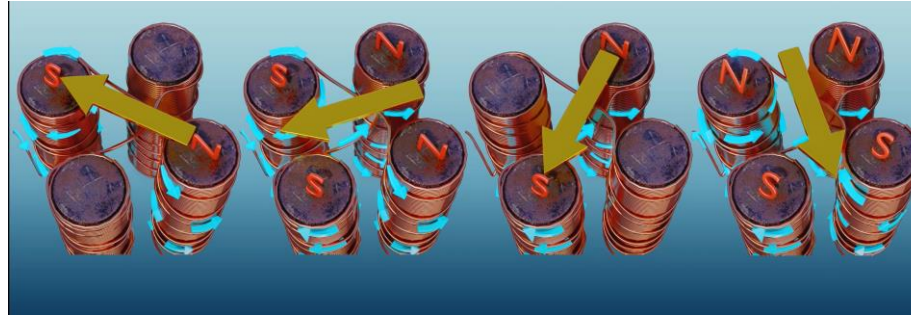


# Rotating Magnetic Field

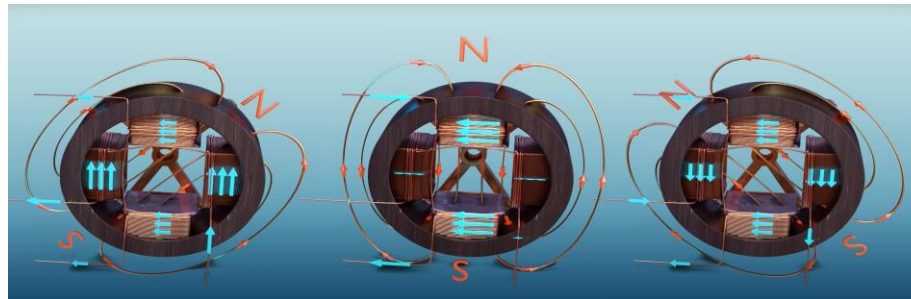
Reza Saeed Kandezy

# How to produce rotating magnetic field ?

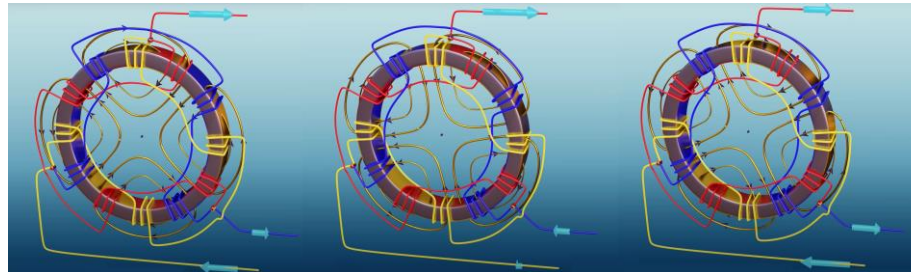
✓ Bailey's  
DC motor



✓ Tesla's  
2-Phase motor

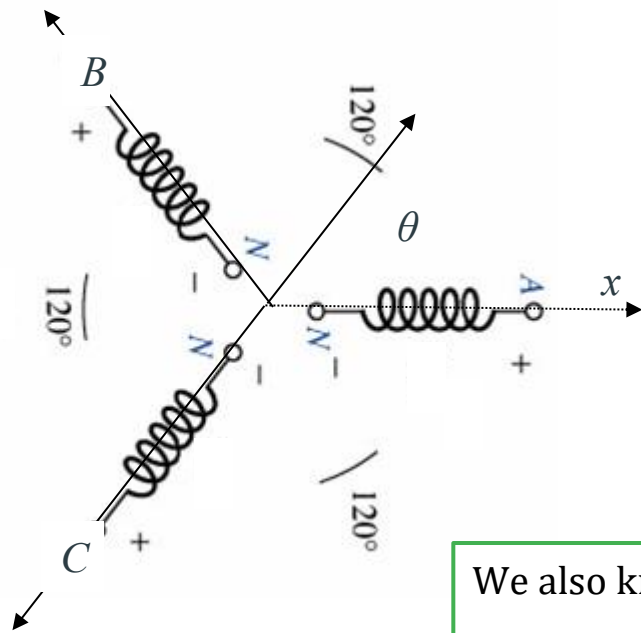


✓ Dobrovolsky's  
3-Phase motor



# Mathematical Interpretation of Rotating M. Field

## 3 phase AC currents & 120 degree apart in space



$$i_a(t) = I_m \cos(\omega t)$$

$$i_b(t) = I_m \cos(\omega t - 120^\circ)$$

$$i_c(t) = I_m \cos(\omega t - 240^\circ)$$

We know that  $B = Ki$

$$B_a = Ki_a(t) \cos(\theta)$$

$$B_b = Ki_b(t) \cos(\theta - 120^\circ)$$

$$B_c = Ki_c(t) \cos(\theta - 240^\circ)$$

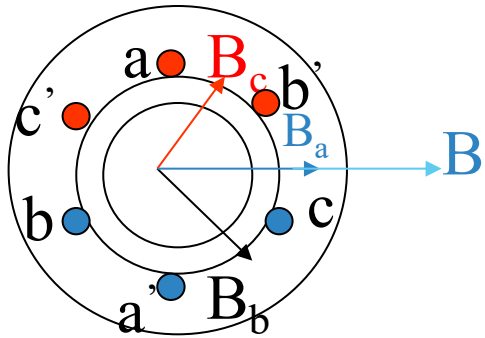
$$B_{\text{total}} = KI_m \cos(\omega t) \cos(\theta) + KI_m \cos(\omega t - 120^\circ) \cos(\theta - 120^\circ) \\ + KI_m \cos(\omega t - 240^\circ) \cos(\theta - 240^\circ)$$

We also know that  $\cos(x) \cos(y) = (1/2)[\cos(x - y) + \cos(x + y)]$

$$B_{\text{total}} = \frac{3}{2} KI_m \cos(\omega t - \theta) + \frac{1}{2} KI_m [\cos(\omega t + \theta) \\ + \cos(\omega t + \theta - 240^\circ) + \cos(\omega t + \theta - 480^\circ)]$$

$$B_{\text{total}} = B_a + B_b + B_c = \frac{3}{2} B_m \cos(\omega t - \theta)$$

# Interpretation of Rotating Magnetic Field

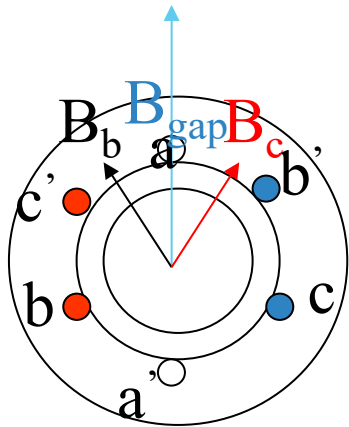


$$B_{\text{gap}} = B_m \cos(\omega t - \theta)$$

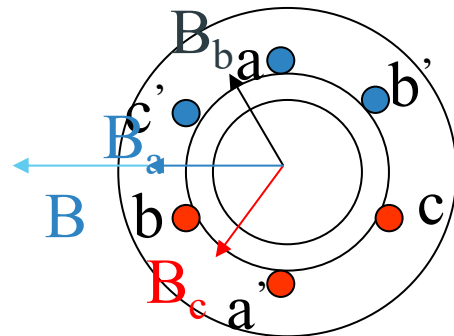
$$B_a(t) = B_m \cos(\omega t)$$

$$B_b(t) = B_m \cos(\omega t - 120^\circ)$$

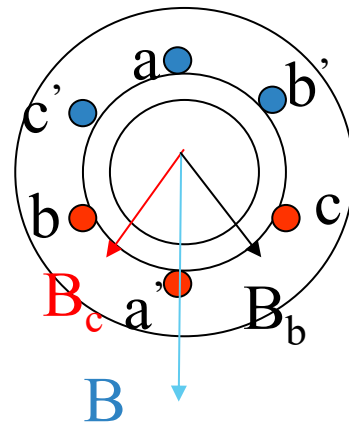
$$B_c(t) = B_m \cos(\omega t - 240^\circ)$$



$$t = t_1, \omega t = 90$$

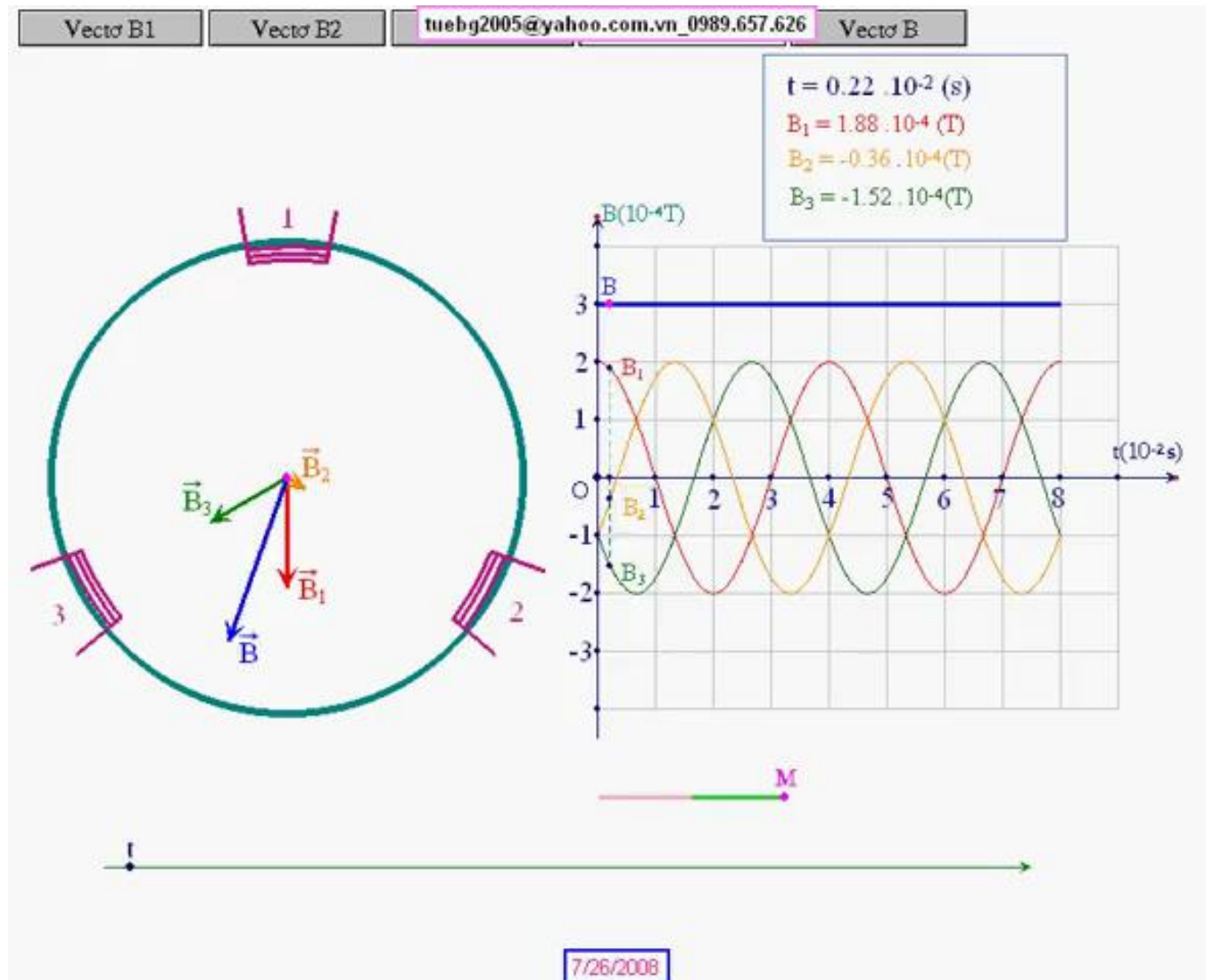


$$t = t_2, \omega t = 180$$



$$t = t_3, \omega t = 270$$

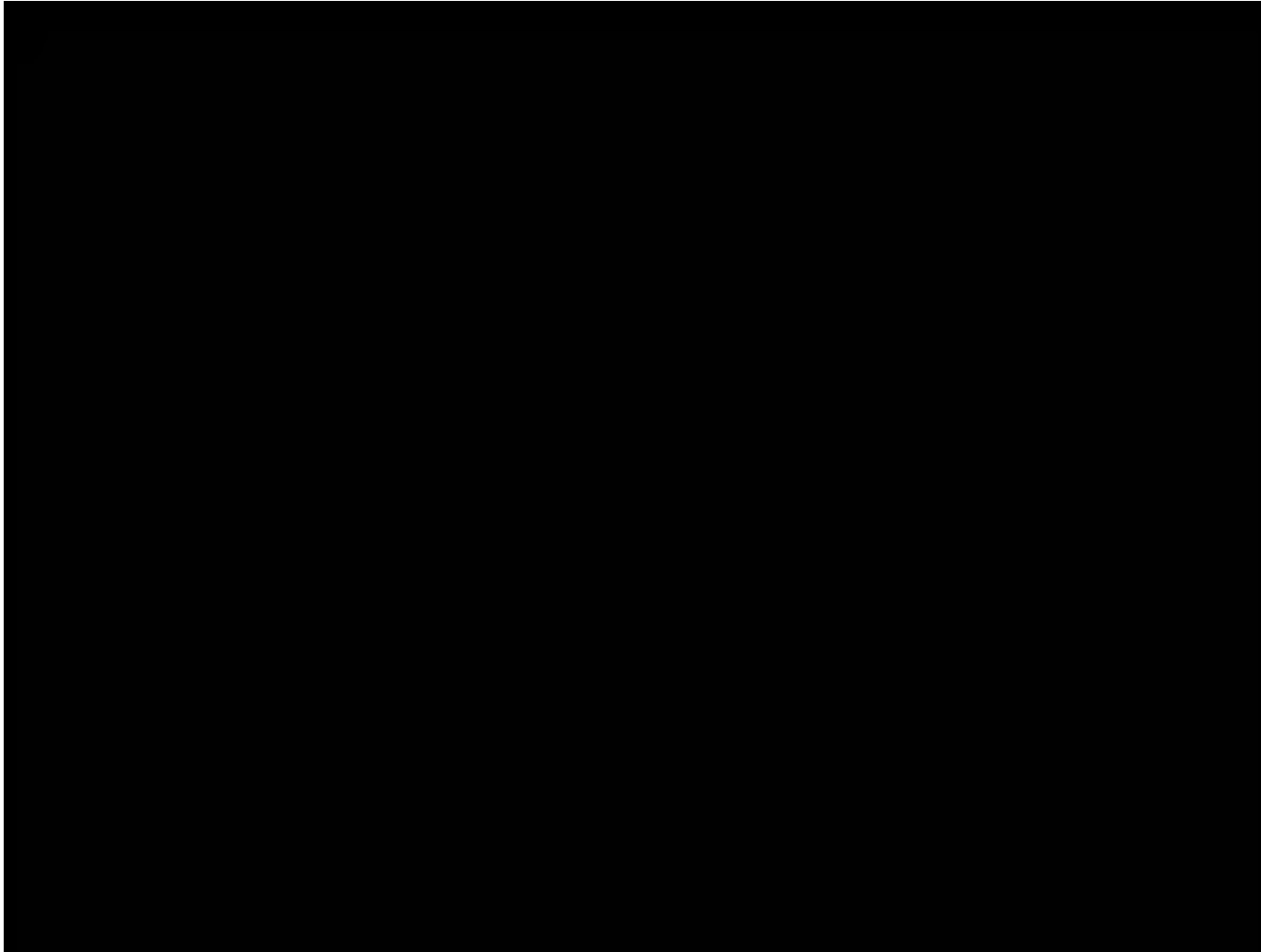
# Interpretation of Rotating Magnetic Field



Let's summarize what we learned about Rotating Magnetic Fields through their application in induction motor



Extra intuitions about rotating magnetic field  
Let's see some fun experiment



Questions

